CutMix-SupCon: Supervised Contrastive Learning with CutMix Augmentation for Multimodal-based Emotion Recognition

서론

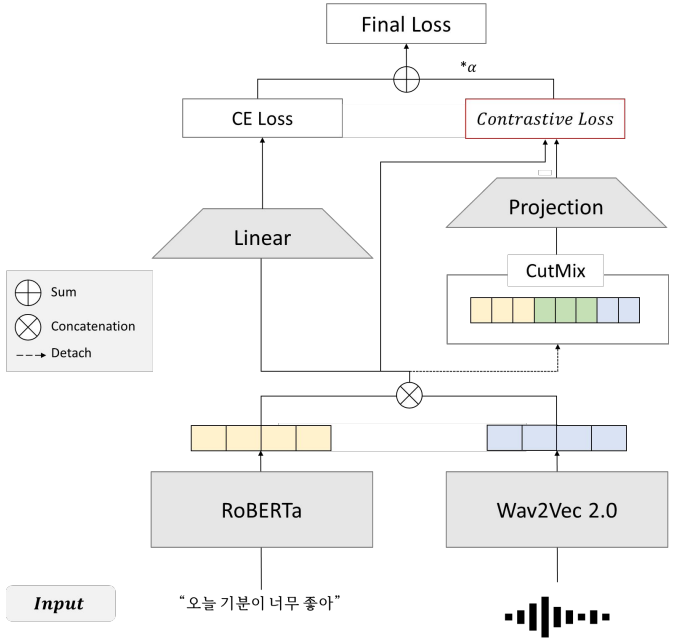
- 멀티모달 데이터는 구조적으로 획득하기 어려운 문제가 있으며, 부족한 데이터 샘플의 문제로 인해 학습된 딥러닝 기반 네트워크의 실용화가 진행되기에는 여전히 한계점이 존재한다.
- 선행 연구들은 실제 환경의 부족한 데이터 문제는 고려하지 않고, 멀티모달 결합 관점에 대해서만 집중되는 경향이 존재한다.
- 본 연구에서는 멀티모달 데이터셋의 불충분한 문제를 해결하기 위해 CutMix-SupCon 방법을 제안한다. 제안된 CutMix-SupCon 방법은 잠재 공간에서 추출된 음성-텍스트 모달 간의 결합된 특징표현으로부터 혼합된 특징표현을 만들고, 혼합 대조 손실을 통해 음성-텍스트의 결합된 멀티모달 특징 표현이 원본과 얼마나 유사한지에 대한 목적 함수를 통해 최적화하였다.

방법

제안하는 멀티모달 CutMix-SupCon 방법은

- 멀티모달 특징표현 결합
- CutMix 기반 멀티모달 혼합 증강
- SupCon

으로 나뉘어지며, 전체 흐름도는 다음과 같다.



- CutMix 기반 멀티모달 혼합 증강

  멀티모달 특징표현 z와 레이블 y에 대하여 샘플 (z_A, y_A), (z_B, y_B) 를 구성할 수 있고, 이로부터 새로운 샘플 $(\tilde{z}, \tilde{y})$를 아래의 식과 같이 만들 수 있다.

$$\tilde{z} = M \odot z_A + (1 - M) \odot z_B$$
$$\tilde{y} = \lambda y_A + (1 - \lambda) y_B$$

- CutMix-SupCon

  CutMix 기법으로부터 증강된 멀티모달 특징표현을 위한 적절한 손실 함수를 적용하기 위해, 잠재 공간내에서 혼합된 정보를 구분하여 효과적인 학습을 가능하게 하는 CutMix-SupCon 방법을 제안한다.

$$L_{CL} = \sum_{i \in I} \frac{1}{|P(i)|} \sum_{p \in P(i)} log \frac{\exp(z_i \cdot \frac{\tilde{z}_p}{\tau})}{\sum_{a \in A(i)} \exp(z_i \cdot \frac{\tilde{z}_a}{\tau})}$$
$$L = L_{CE} + \alpha L_{CL}$$

  위와 같은 최종 손실 함수를 통해 모델은 혼합된 멀티모달 특징표현이 원본의 샘플의 고유 정보를 얼마나 포함하는지를 학습한다.

실험 및 결과

- 데이터셋: KEMDy19 Text+Audio 멀티모달 감정인식 데이터셋

  중복된 파일, 음성의 발화 세그먼트, 20초 이상의 음성 파일 제거 등의 전처리를 통해 10,214개의 데이터 샘플을 확보

  학습 및 평가셋 8:2 무작위 분할
- 방법: 각 모달에 대한 특징표현 기반의 CE, Mixup, CutMix, 제안하는 Cutmix-SupCon 방법에 대한 결과 보고

Input	Method	Acc (%)	F1-score (%)
Text	Cross-Entropy	73.42	69.32
Audio		72.59	67.13
Text+Audio		73.57	69.08
Text	Mixup	71.76	67.37
Audio		71.46	65.89
Text+Audio		72.39	68.94
Text	CutMix	72.93	69.37
Audio		71.76	65.94
Text+Audio		74.74	70.77
Text	CutMix-SupCon (ours)	72.49	**69.72**
Audio		72.54	**68.27**
Text+Audio		75.28	**71.17**

표 1 KEMDy19 데이터셋을 사용한 감정인식 평가 결과

Input	Method	Weighted F1-score(%)
Text+Audio	Cross-Entropy	73.55
	Mixup	72.39
	CutMix	74.77
	CutMix-SupCon (ours)	**74.96**

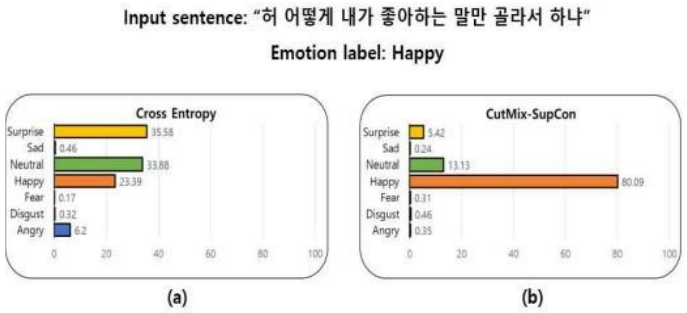


그림 2 KEMDy19 테스트 데이터 샘플에 대한 CE와 CutMix-SupCon의 예측 확률 분포 비교

Adversarial Fine-tuning using Generated Respiratory Sound to Address Class Imbalance

June-Woo Kim [1,4], Chihyeon Yoon [1], Miika Toikkanen [2], Sangmin Bae [3,4], Ho-Young Jung [1†]

[1] Kyungpook National University, [2] ALI Co., Ltd, [3] KAIST AI [4] RSC LAB, MODULABS

{kaen2891, chichi8969, hoyjung}@knu.ac.kr, miika.toikkanen.2@gmail.com, bsmn0223@kaist.ac.kr

Introduction

- While ***deep generative models*** have emerged as a promising approach in the medical image domain to address data scarcity, their use for sequential data like **respiratory sounds** is less explored.
- In this work, we propose a straightforward approach to augment imbalanced respiratory sound data using an **audio diffusion model** as a ***conditional neural vocoder***.
- Nevertheless, the **discrepancy between synthetic and real samples** can introduce problems related to ***distribution inconsistency***, which can degrade the performance of respiratory sound classification models as the proportion of synthetic data in the training set increases.
- We further introduce a simple yet effective **Adversarial Fine-tuning** method to ***align features between the synthetic and real respiratory sound samples*** to improve respiratory sound classification performance.

Method 1

- We aim to generate ***high-fidelity respiratory sound samples*** using audio diffusion model [1] and then **combine these synthetic samples with real data to improve the respiratory sound classification task**, especially for imbalanced lung sound disease classes.
- In phase 1, we introduce a simple method for **generating respiratory sound samples** using a ***conditional neural vocoder*** inspired by the recent success of ***DiffWave*** [1] in obtaining realistic audio.

Phase 1: Respiratory Sound Generation

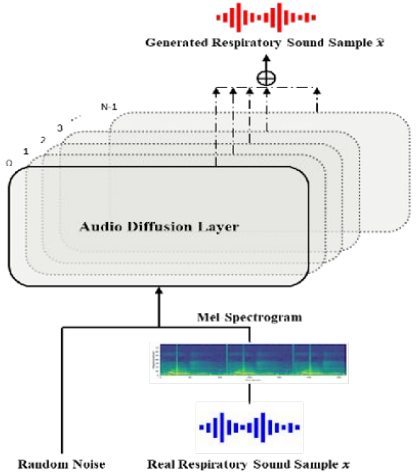


Figure 1: Overview of respiratory sound samples generation using audio diffusion model.

Method 2

- While augmenting real data with synthetic samples can be beneficial, we found in early experiments that in our case the **distribution mismatch between the two types of data degraded the performance of the classification model.**
- To overcome this issue, we propose a simple yet effective ***Adversarial Fine-Tuning (AFT)*** method inspired by [2].

Phase 2: Adversarial Fine-tuning with Mixed Samples for Respiratory Sound Classification

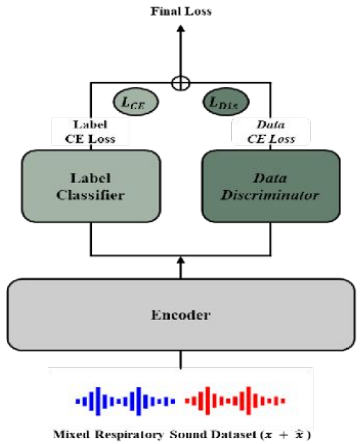


Figure 2: Overview of Adversarial fine-tuning with synthetic and real samples.

- The proposed method consists of two losses with label classifier $\mathcal{L}_{\text{CE}}$ and data discriminator $\mathcal{L}_{\text{Dis}}$:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{n} y_i \log(\hat{y}_i), \qquad \mathcal{L}_{\text{Dis}} = -\frac{1}{N}\sum_{i=1}^{n} d_i \log(\hat{d}_i),$$

- where $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{Dis}}$ are ***CE loss*** with label y and label d, and the predicted probabilities $\hat{y}$ and $\hat{d}$ are obtained by passing through the ***classifier*** and ***discriminator***, respectively.
- To ensure that the *learned features cannot distinguish between the synthetic and real samples*, **gradients from $\mathcal{L}_{\text{Dis}}$ are multiplied by a negative constant during the backpropagation.**
- The final training objective is $\mathcal{L}_{\text{Final}} = \mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{Dis}}$ where λ is a regularization parameter drawn from [2].
- We used Audio Spectrogram Transformer (AST) [3] model as our baseline.

Results

Table 1: Performance according to various mixed sample amounts.

method	training dataset							
	No Aug.	Mixed-500	Mixed-800	Mixed-1k	Mixed-1.5k	Mixed-2k	Mixed-3k	Mixed-5k
AST FT	59.55	$59.92_{\pm 0.82}$	$59.99_{\pm 1.17}$	$59.81_{\pm 0.36}$	$59.65_{\pm 0.30}$	$59.18_{\pm 0.65}$	$59.04_{\pm 0.32}$	$58.56_{\pm 0.84}$
AFT	-	**$61.79_{\pm 0.47}$**	$60.89_{\pm 0.78}$	$60.8_{\pm 1.05}$	$60.03_{\pm 1.14}$	$60.64_{\pm 0.45}$	$59.96_{\pm 0.38}$	$59.74_{\pm 0.6}$

Table 2: Accuracy (%) of the abnormal class on the ICBHI test set for AST FT and AFT.

class	ratio	method (dataset)				
		AST FT (No aug.)	AST FT (Mixed-500)	AST FT (Mixed-2k)	Adversarial FT (Mixed-500)	Adversarial FT (Mixed-2k)
crackle (C_c)	23.55%	45.45	42.84	42.86	44.07	**46.99**
wheeze (C_w)	13.97%	36.62	36.1	22.08	**37.92**	30.12
both (C_b)	5.19%	15.38	9.09	7.69	**41.96**	35.66

Table 3: Overall comparison of the ICBHI dataset for the respiratory sound classification task.

method	architecture	pretrain	S_p (%)	S_e (%)	Score (%)
RespireNet [8] (CBA+BRC+FT)	ResNet34	IN	72.30	40.10	56.20
Wang *et al.* [32] (Splice)	ResNeSt	IN	70.40	40.20	55.30
Nguyen *et al.* [25] (CoTuning)	ResNet50	IN	79.34	37.24	58.29
Moummad *et al.* [23] (SCL)	CNN6	AS	75.95	39.15	57.55
Bae *et al.* [3] (Fine-tuning)	AST	IN + AS	77.14	41.97	59.55
Bae *et al.* [3] (Patch-Mix CL)	AST	IN + AS	**81.66**	**43.07**	**62.37***
AFT on Mixed-500 [ours]	AST	IN + AS	$80.72_{\pm 0.99}$	$42.86_{\pm 1.3}$	$61.79_{\pm 0.47}$

- While directly fine-tuning on mixed data did not improve the performance of the minority classes overall, **our proposed method improved their accuracies by up to 26.58%, especially in C_b.**
- Furthermore, our proposed method trained with ***Mixed-500*** achieved a Score of **61.79%, outperforming the AST FT model by 2.24%.**

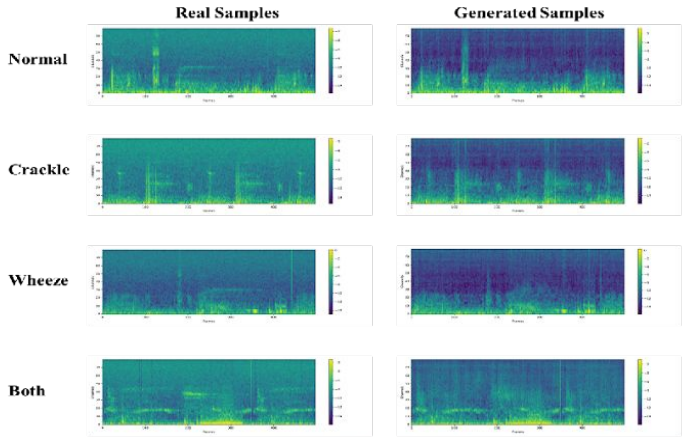


Figure 3: Comparison of spectrograms per each class randomly chosen from the test set and the generated results.

Generated Samples

Github Code

Paper Links

References

[1] Kong Zhifeng, et al (2021). "DiffWave: A versatile Diffusion Model for Audio Synthesis." *ICLR* 2021

[2] Ganin Yaroslav, et al (2016). "Domain-Adversarial Training of Neural Networks.". The Journal of Machine Learning Research 2016

[3] Gong Yuan, et al (2021). "AST: Audio Spectrogram Transformer". *INTERSPEECH* 2023

A Military Audio Dataset for Situational Awareness and Surveillance

Introduction

- Audio classification related to military activities is a challenging task due to the high levels of background noise and the lack of suitable and publicly available datasets.
- To bridge this gap, this paper constructs and introduces a new military audio dataset, named MAD, which is suitable for training and evaluating audio classification systems.
- The proposed MAD dataset is extracted from various military videos and contains 8,075 sound samples from 7 classes corresponding to approximately 12 hours, exhibiting distinctive characteristics not presented in academic datasets typically used for machine learning research.
- We further conduct a comprehensive sound classification study of various deep learning algorithms on the MAD dataset.

Method

- Methodology used to collect the MAD dataset

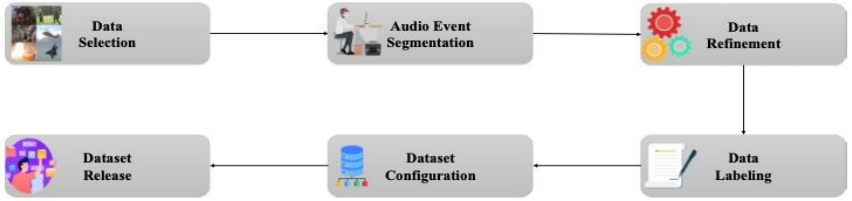


Fig. 1 Illustration of MAD dataset collection procedure.

- Data selection
  Identify the audio content types for the seven military-related categories.
- Audio event segmentation
  Identify the start and end times of each audio event.
- Data refinement
  Refine the data using a variety of techniques, such as formatting, resampling, and extracting.
- Data labeling
  Label all audio events into one of the seven predefined classes.
- Data configuration
  Randomly split all the data into train/test sets with an approximately 9:1 ratio.

- Dataset Overview
  - Total Data: 8,075 audio events (12.05 hours in total)
  - Dataset Split
    Training Set: 7,393 samples (10.98 hours)
    Test Set: 682 samples (1.07 hours)

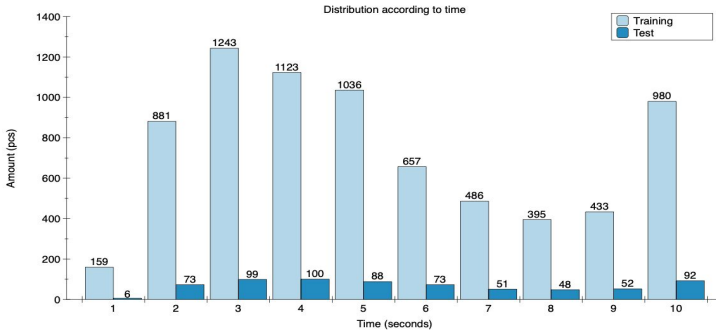


Fig. 3 Bar chart illustrating the distribution of training and test sets according to various times.

- Key Feautres
  All audio events range from 1 to 10 seconds in duration.
  Meticulously labeled and categorized for high-quality deep learning-based audio classification tasks.
- Total Classes: 7 audio classes, including Communication, Gunshot, Footsteps, Shelling, Vehicle, Helicopter, and Fighter.

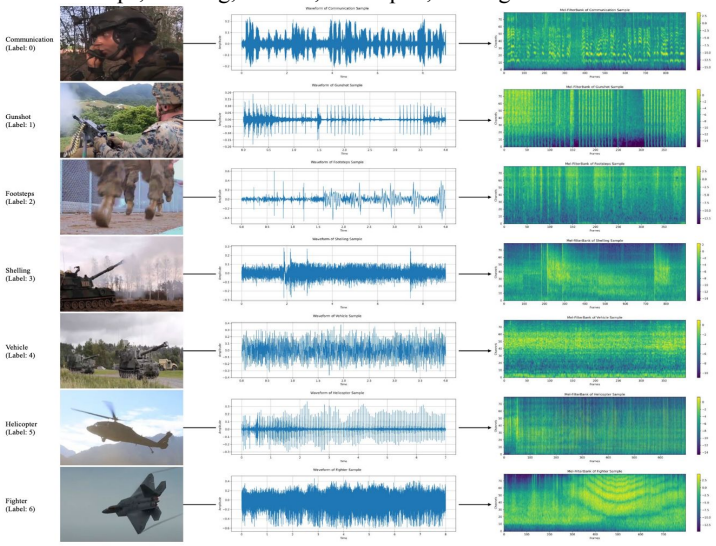


Fig. 4 The overall pre-processing task for converting waveform to Mel filterbank.

Technical Validation

- Model Evaluation: We conducted a comprehensive analysis of various deep learning models to evaluate the performance of the MAD dataset.
  - Models: ResNet, EfficientNet, CNN-N (PANNs), and AST (Audio Spectrogram Transformer).
- Pre-processing
- Audio waveforms were converted into Log Mel filterbank features (80 dimensions, 25ms window, 10ms overlap).
- All audio samples were standardized to a 10-second length using zero padding.
- Standard normalization was applied to the spectrograms.
- Augmentation: SpecAugment was used to mitigate overfitting.

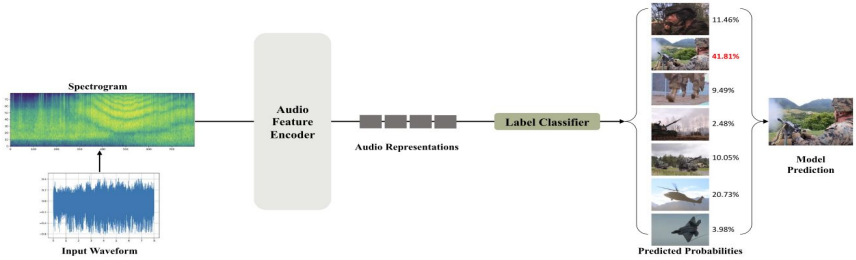
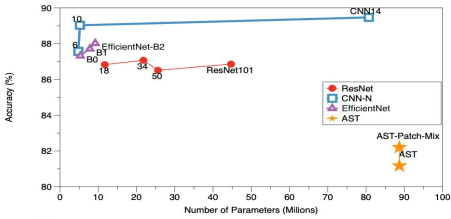
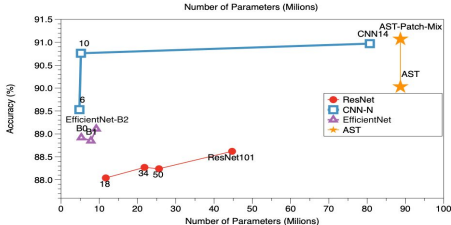


Fig. 5 Illustration of overall deep learning-based audio classification process.

- Experimental Result Summary
- Pretrained models consistently outperformed those trained from scratch. With pretrained weights, the AST-Patch-Mix model achieved the highest accuracy (91.07%), aligning with the trend that models with more parameters perform better. A notable exception was the CNN10 model, which achieved a strong 89.53% accuracy with only 6% of the parameters of the AST model.
- In contrast, when trained from scratch, the CNN14 model performed best (89.47%), while the high-parameter AST models surprisingly showed the lowest performance.

architecture	parameters	pretrain	accuracy (%)	
			pretrained weights	scratch
ResNet18	11.7M		$88.04_{\pm 0.59}$	$86.83_{\pm 0.38}$
ResNet34	21.8M	ImageNet	$88.27_{\pm 0.29}$	$87.07_{\pm 0.74}$
ResNet50	25.6M		$88.24_{\pm 0.41}$	$86.51_{\pm 0.38}$
ResNet101	44.7M		$88.62_{\pm 0.31}$	$86.86_{\pm 0.60}$
EfficientNet-B0	5.3M		$88.92_{\pm 0.17}$	$87.32_{\pm 1.00}$
EfficientNet-B1	7.8M	ImageNet	$88.85_{\pm 0.28}$	$87.71_{\pm 0.33}$
EfficientNet-B2	9.2M		$89.11_{\pm 0.64}$	$88.02_{\pm 0.27}$
CNN6	4.8M		$89.53_{\pm 0.26}$	$87.57_{\pm 0.81}$
CNN10	5.2M	AudioSet	$90.76_{\pm 0.47}$	$89.03_{\pm 0.37}$
CNN14	80.7M		$90.97_{\pm 0.34}$	$\mathbf{89.47}_{\pm 0.22}$
AST	87.7M	ImageNet + AudioSet	$90.03_{\pm 0.23}$	$81.17_{\pm 0.77}$
AST-Patch-Mix		ImageNet + AudioSet	$\mathbf{91.07}_{\pm 0.19}$	$82.20_{\pm 1.13}$





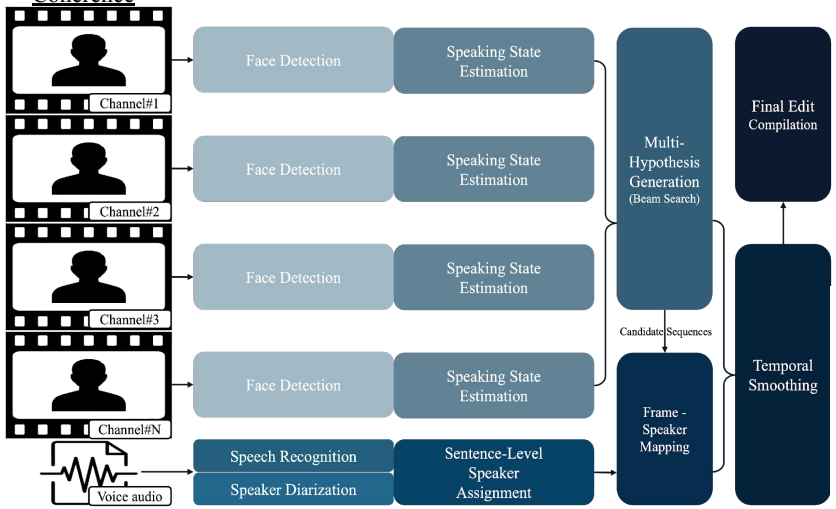
Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis Over Multi-Channel Sources

Introduction

- An automated system for editing multi-channel video by synchronizing raw inputs through separate visual and auditory pipelines.
- Deep late-fusion of multimodal information for face recognition, speaking-state estimation, speech recognition and speaker identification.
- A multi-hypothesis editing approach that provides candidate versions based on probability distribution over time for extracting frames from each channel stream.
- Frame-level match accuracies of 84.99% and 90.10% for the 3-channel and 5 channel videos, highlighting the system's robust performance across varying speaker complexities.

Method

- For multi-speaker, multi-channel content, we adopted an intuitive approach to make automated editing feasible. This is primarily because datasets containing multi-speaker, multi-channel videos paired with their fully edited versions are extremely rare.
- The lack of such datasets makes it challenging to apply standard supervised learning techniques effectively for automated video editing.
- For the audio modality, our goal is to accurately detect active speech segments and identify the corresponding speakers from diverse and often noisy multi-channel sources. - Speech Recognition, Speaker Diarization
- The visual approach involves estimating the speaking state based on the facial region of the speaker. - Face Detection, Speaking State Estimation
- Integration and Channel Selection - Multi-Hypothesis Generation for Temporal Coherence



Experiment & Result

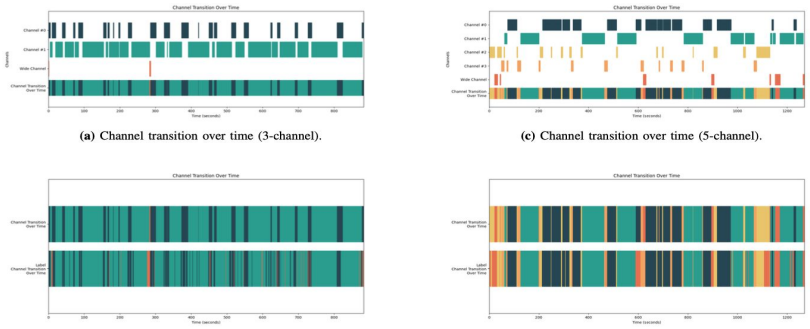
- Dataset
  - Studio recordings provided by a Korean public broadcasting company in South Korea. {3-channel video dataset, 5-channel video dataset}
- Evaluation
  - Compare the ground-truth labeled data with output generated by our system at the frame level.
  - For the Quantitative Analysis, Analyze the frame-level alignment accuracy under various smoothing and temporal conditions.

Data	Thr 15	Thr 30	Thr 45
Best Path (self-transition bias 0.7)			
3-channel video	0.8498	0.8495	0.8495
5-channel video	0.8995	0.8999	0.8854
Best Path (self-transition bias 0.9)			
3-channel video	**0.8499**	0.8497	0.8481
5-channel video	0.8987	**0.9010**	0.8865

TABLE I: Match rates (Top Beam Paths applying self-transition bias 0.7/0.9) varying the smooth threshold.

Length (s)	3-channel video			5-channel video		
	Match	Frames	Rate	Match	Frames	Rate
0–5	3,849	6,660	0.5779	2,981	4,335	0.6877
5–10	4,162	4,695	0.8865	5,447	5,940	0.9170
10–20	5,161	5,340	0.9665	2,867	4,425	0.6479
20–30	2,971	3,210	0.9255	3,573	3,720	0.9605
30–40	1,140	1,140	1.0000	8,412	8,580	0.9804
40–50	–	–	0.0000	2,524	2,550	0.9898
50–60	3,104	3,315	0.9363	1,648	1,650	0.9988
60–70	–	–	0.0000	2,010	2,010	1.0000
70–80	2,128	2,130	0.9991	2,198	2,220	0.9901
80–90	–	–	0.0000	2,670	2,670	1.0000
Total	22,515	26,490	**0.8499**	34,330	38,100	**0.9010**

TABLE II: Match rates across different segment lengths.



(a) Channel transition over time (3-channel). (c) Channel transition over time (5-channel).

(b) Transition and labels across segments (3-channel). (d) Transition and labels across segments (5-channel).

Fig. 2: Visualization of channel selection for the 3-channel (left column) and 5-channel (right column) videos. Each sub-figure shows the overall transitions and a detailed view with ground-truth labels.

- For the Qualitative Analysis, Analyze subjective assessments of editing quality and viewer experience.
- The analysis was performed with 24 participants, including developers and broadcasting industry professionals, under a blind setup, where they were not informed whether a video was edited by our system or by a human expert.

	3-channel	5-channel	
Criteria	Ours	Ours	Expert
1. Speaker Switching Smoothness & Seamlessness			
1.1 Is the camera transition smooth when the speaker changes in the video?	4.26±0.81	4.43±0.78	3.95±1.02
1.2 Are the cut transitions in the video not awkward?	4.26±1.00	4.34±0.64	4.04±1.02
2. Scene Consistency			
2.1 Are the scenes in the video consistently focused on the speaker in context?	4.17±0.88	4.52±0.59	3.95±0.92
3. AI Editing Potential			
Compare the AI-edited video with professional editing videos and determine			
3.1 if the AI-edited video can replace professional editing videos?	–	4.08±1.08	
3.2 if the AI automatic editing system can help professional editors?	–	4.56±0.66	

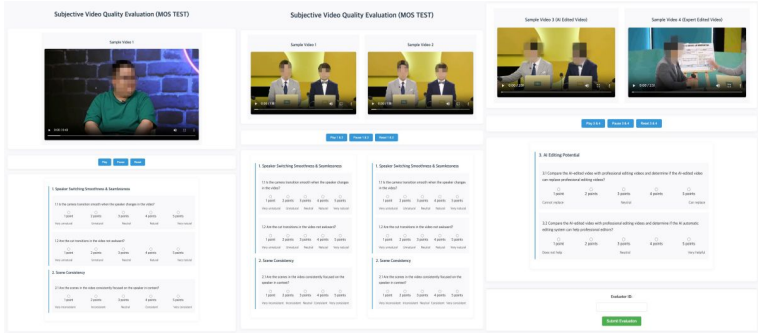
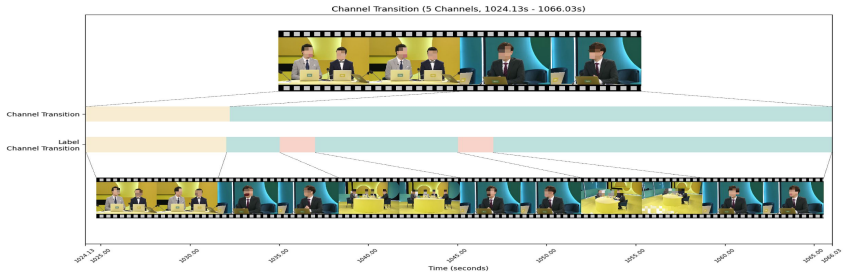



TABLE III: Subjective Video Quality Evaluation Results.